

HR Attrition Prediction Project - Complete Explanation

1. Project Objective

The goal of the project is to predict whether an employee will leave the company or stay. This helps HR departments identify high-risk employees early and improve retention strategies.

2. Importing Libraries

Libraries such as pandas, numpy, matplotlib, and seaborn were imported. These libraries help in: - Data handling - Numerical operations - Visualization - Machine learning workflow support
Memory Trick: Libraries are the toolbox of machine learning.

3. Loading Dataset

The dataset was loaded using `pandas.read_csv()`. The dataset contains employee information such as: - Salary - Age - Department - Overtime - Attrition
Purpose: To bring employee records into memory for processing.
Memory Trick: Opening the HR Excel sheet.

4. Data Understanding

Functions used: - `head()` - `info()` - `describe()` - `isnull()`
Purpose: - Understand columns - Detect missing values - Check datatypes - Understand distribution
Memory Trick: Doctor checking patient before treatment.

5. Duplicate Removal

`drop_duplicates()` was used because duplicate employee records can bias learning. 97 duplicate rows were removed.
Memory Trick: Same student should not write exam twice.

6. Missing Value Handling

`SimpleImputer(strategy='mean')` was used. Purpose: Machine learning models cannot process NaN values. Why mean? Mean preserves overall distribution. Alternatives: - Median - KNN Imputer - Row deletion
Memory Trick: Fill missing answers intelligently.

7. Label Encoding

LabelEncoder() converts text into numerical values. Example: Yes → 1 No → 0 Why used? Machine learning models require numerical input. Alternative: One Hot Encoding. Memory Trick: Translate English into machine language.

8. Removing Constant and ID Columns

Columns like EmployeeNumber and constant columns were removed. Reason: They provide no predictive information. Memory Trick: Roll number does not predict marks.

9. Heatmap

sns.heatmap(df.corr()) visualized feature relationships. Purpose: Understand which features are strongly related. Memory Trick: Friendship map of features.

10. Train-Test Split

train_test_split(test_size=0.2, random_state=42) Purpose: Separate training and testing data. Hyperparameters: - test_size=0.2 → 20% testing data - random_state=42 → reproducible results Memory Trick: Practice set vs final exam.

11. StandardScaler

StandardScaler() normalizes feature values. Purpose: Prevent large-value features from dominating. Important for: - KNN - Logistic Regression Memory Trick: Equal shoes for all features.

12. Logistic Regression

LogisticRegression(max_iter=1000) Purpose: Baseline binary classifier. Hyperparameter: max_iter=1000 prevents convergence failure. Output: Good baseline accuracy but lower recall. Memory Trick: Beginner classifier.

13. Decision Tree

DecisionTreeClassifier(criterion='gini', max_depth=5) Purpose: Rule-based learning. Hyperparameters: - criterion='gini' → impurity measurement - max_depth=5 → prevents overfitting Memory Trick: Interview flowchart.

14. Random Forest

RandomForestClassifier(n_estimators=100) Purpose: Ensemble learning using multiple decision trees. Hyperparameter: n_estimators=100 → number of trees. Why better? Reduces overfitting and handles nonlinear relationships. Memory Trick: Committee of experts.

15. KNN

KNeighborsClassifier(n_neighbors=5) Purpose: Similarity-based prediction. Hyperparameter: n_neighbors=5 → uses 5 nearest employees. Memory Trick: Employees behave like neighbors.

16. XGBoost

XGBClassifier(n_estimators=100, eval_metric='logloss') Purpose: Advanced boosting algorithm. Hyperparameters: - n_estimators=100 → boosting rounds - eval_metric='logloss' → classification error metric Memory Trick: Self-improving learner.

17. Evaluation Metrics

Metrics used: - Accuracy - Precision - Recall - F1 Score - ROC-AUC Most important: Recall because missing attrition cases is dangerous. Memory Trick: Model report card.

18. Confusion Matrix

Confusion matrix shows: - TP - TN - FP - FN Purpose: Detailed error analysis. Memory Trick: Prediction report card.

19. Feature Importance

rf.feature_importances_ was used. Purpose: Identify which employee features affect attrition most. Calculated using impurity reduction. Memory Trick: Top suspects list.

20. SMOTE

SMOTE(random_state=42) Purpose: Balance minority class. Before SMOTE: More employees stayed than left. After SMOTE: Balanced classes improved recall. Memory Trick: Give minority class equal voice.

21. Final Conclusion

The project successfully built a complete HR attrition prediction pipeline. After preprocessing, scaling, encoding, model training, evaluation, and balancing using SMOTE, Random Forest and XGBoost performed best because ensemble methods captured complex employee behavior patterns effectively.